

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Data Analysis and Visualization	Course Code:	DS3001
	Degree Program:	BS (Data Science)	Semester:	Fall 2023
	Exam Duration:	60 Minutes	Total Marks:	50
	Paper Date:	11 th -November-2023	Weight	15%
	Section:	ALL	Page(s):	7 + 1 (Rough Page)
	Exam Type:	Mid-2		

Name: _____ Roll No. _____ Section: _____

- Instruction/Notes:**
- Attempt all questions on the provided question paper.
 - Space for rough work is provided at the end of the paper.
 - Show all calculations, direct answers are not acceptable.
 - Even if you do use extra rough sheets, they should **NOT** be attached with final paper.

Question #	1	2	3	Total
Total Marks	20	10	20	50
Obtained Marks				

Question 1:

We want to train the logistic regression model for classification problem. You are provided with a text passage for training purpose.

Input	Actual Label
What truly sets this book apart is the depth of emotion it evokes. I laughed, I cried, and I felt my heart race with anticipation during the most gripping moments. The themes explored in this story are both timely and timeless, touching on the complexities of human nature, the power of friendship, and the triumph of hope in the face of adversity. I cannot recommend this book enough. It is a masterpiece of modern literature that deserves a place on every bookshelf. Whether you're a seasoned reader or just looking for a captivating story to dive into, this book will not disappoint. Prepare to be transported on an unforgettable journey that will stay with you for a lifetime.	1

Each training observation would be represented by the 5 crafted features shown in the following table.

Features	
1	log of Word count
2	number of punctuations (period, comma, apostrophe, quotation, question, exclamation, colon etc.)
3	number of positive words
4	number of negative words
5	ratio of capitalized words (words starting with capital letter) to total words

Dictionary for positive and negative words is given below

Positive dictionary	Negative dictionary
captivating, depth, enjoy, friendship, good, gripping, happy, laughed, lifetime, masterpiece, modern, proud, triumph, timeless, unforgettable, unforgettable	adversity, anger, anticipation, bad, cannot, complexities, cried, disappoint, hopeless, not, rude, sad, seasoned, waste, weird

Your task is to update the weights once, using stochastic gradient descent algorithm. Assume all initial weights set to 0.2 and learning rate $\alpha=0.5$. Also compute the Loss function: binary cross entropy loss.

Features	X_1	X_2	X_3	X_4	X_5
Values					

Logistic regression Model:

Actual Label y : _____ , **Predicted value $\hat{y}$:** _____

Loss function:

Weight update equations:

Question 2:

Suppose we have collected the following Bigram and Unigram frequencies from a text corpus. Also assume that all other observed counts are 0. The total number of words in our vocabulary $V=24$.

	computer	keyboard	monitor	store
computer	0	2	4	4
keyboard	1	0	0	1
monitor	0	1	1	1
store	2	0	0	0

Bigram Frequencies

computer	keyboard	monitor	store
10	3	5	6

Unigram frequencies

Consider the following sentence fragment S: "I shopped at the computer _____"

You need to determine whether the sentence is more likely to end with "computer store" or "computer monitor".

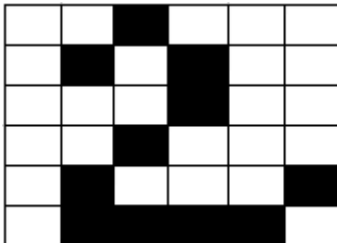
- Compute the **raw bigram probabilities** for the candidate words {store, monitor} to complete the sentence S, i.e. $P(\text{store}|\text{computer})$ and $P(\text{monitor}|\text{computer})$. Is one word more likely than the other, and if so which one?
- Compute the **smoothed bigram probability** for the candidate words {store, monitor} using Add-one smoothing. Is one word more likely than the other, and if so which one?

Question 3:

Consider the pre-trained CNN stack given below. It is trained on 2 classes: symbol (0), digit (1).

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For the given image, you need to determine what class label would be assigned to it. Show all intermediate setps.
 Black pixel=1, white pixel=0.



The output of Conv-Layer (Show all steps):

Output of MaxPool Layer:

Output of Fully Connected Layer (Show complete working):

Class label of the input image:

Rough Work